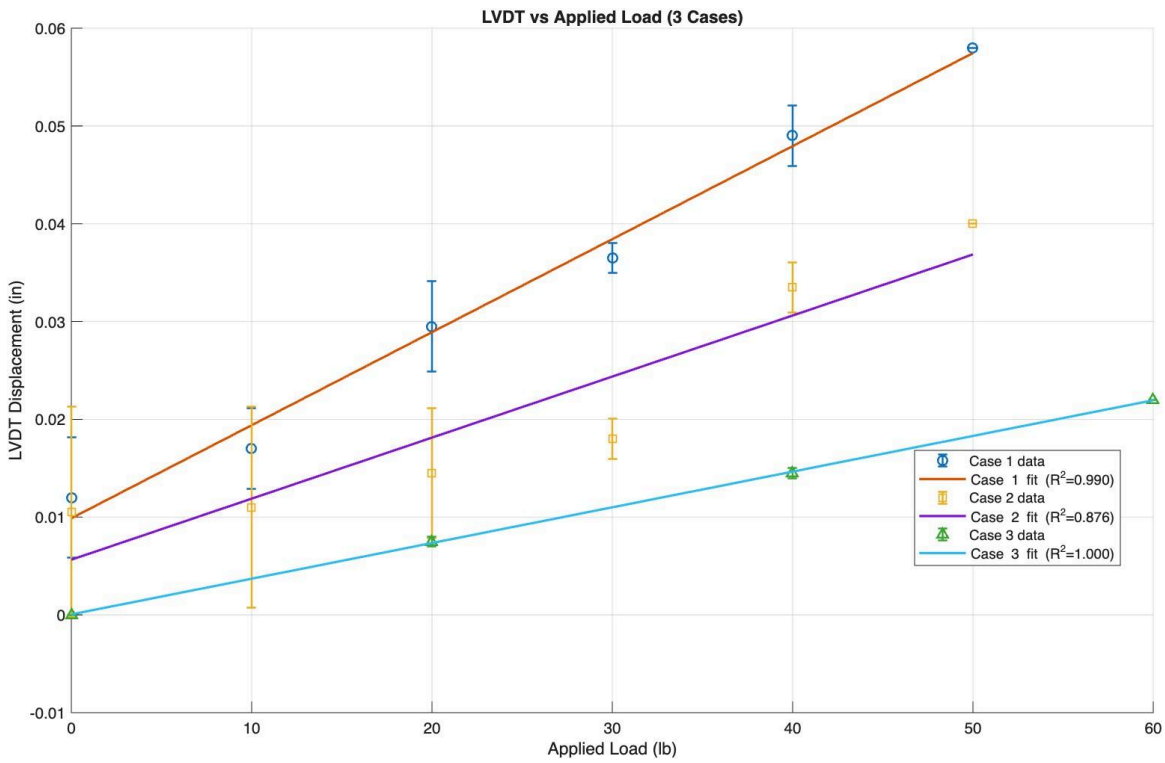
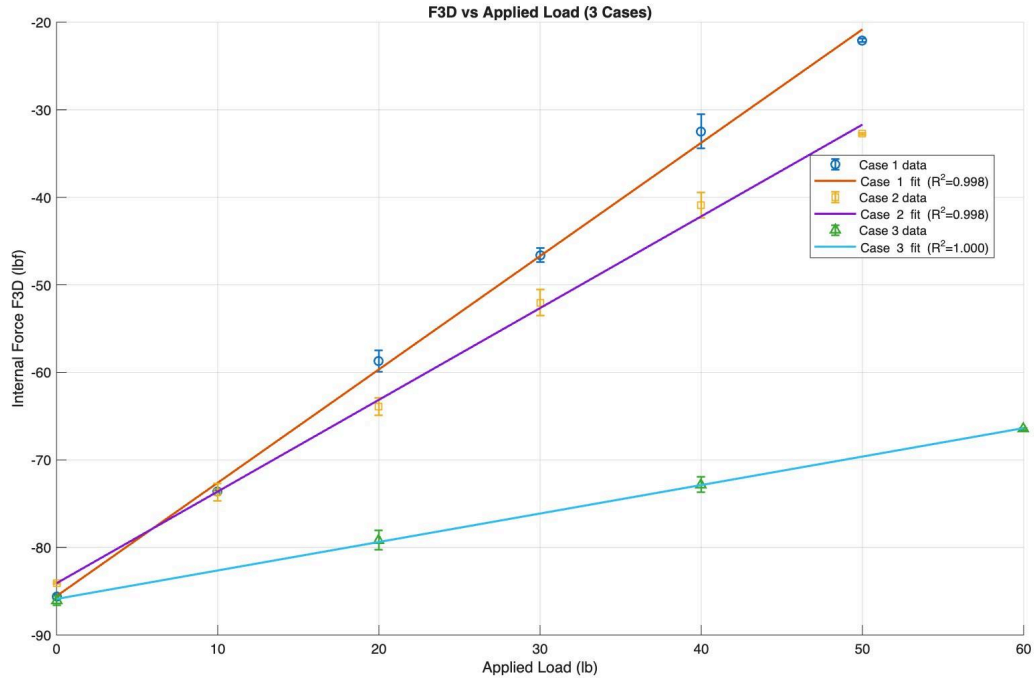
ASEN 3802 Lab 1 Part 2: Structural Mechanics (Truss Lab)

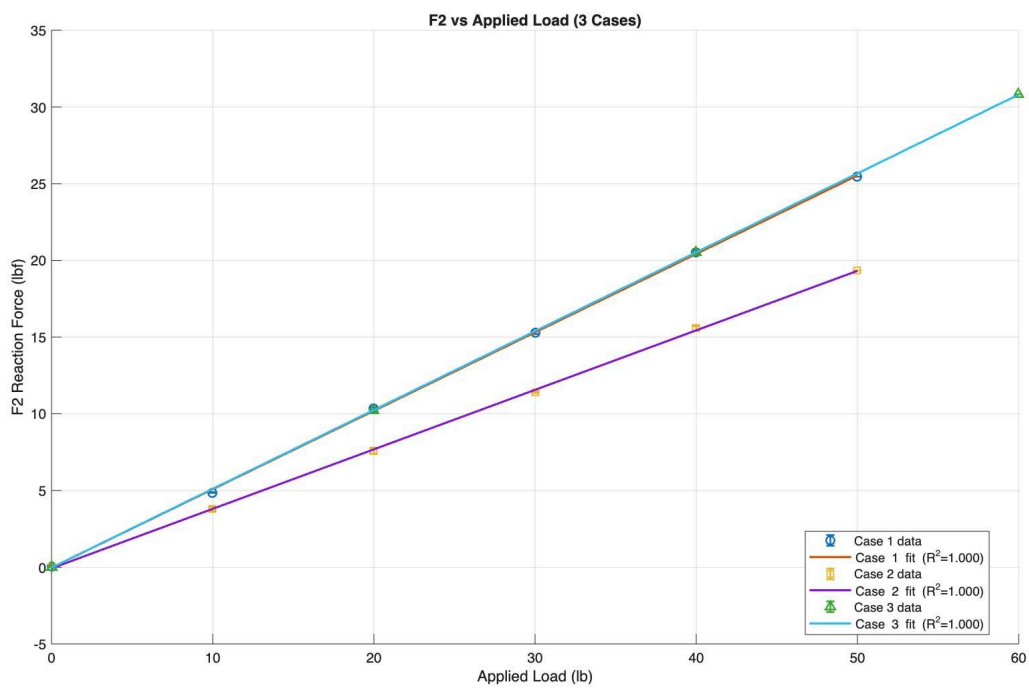
Quintin Jochim, Anton Pylypchenko, Ritik Sarraf, Harroop Sooch

Smead Aerospace CU Boulder ASEN 3802: Aerospace Laboratory

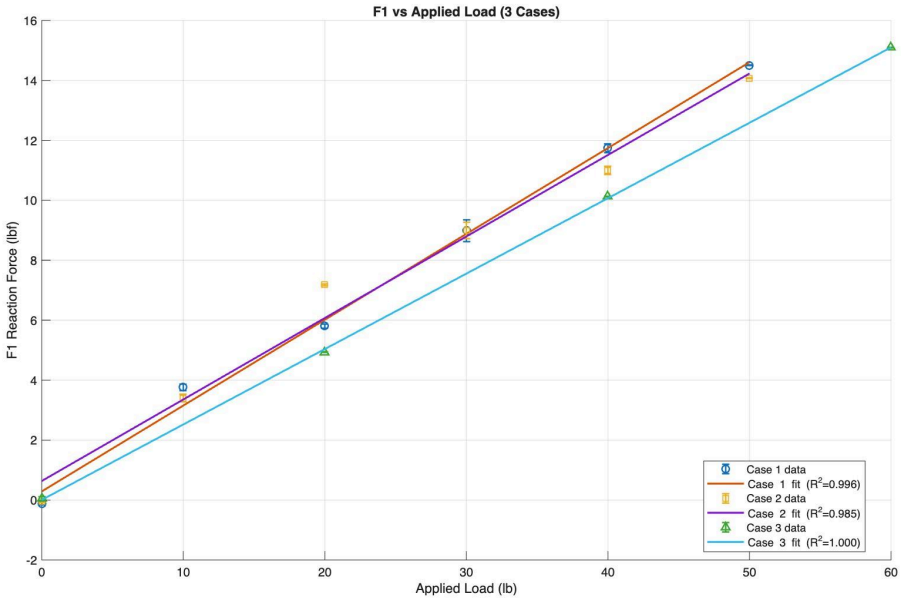


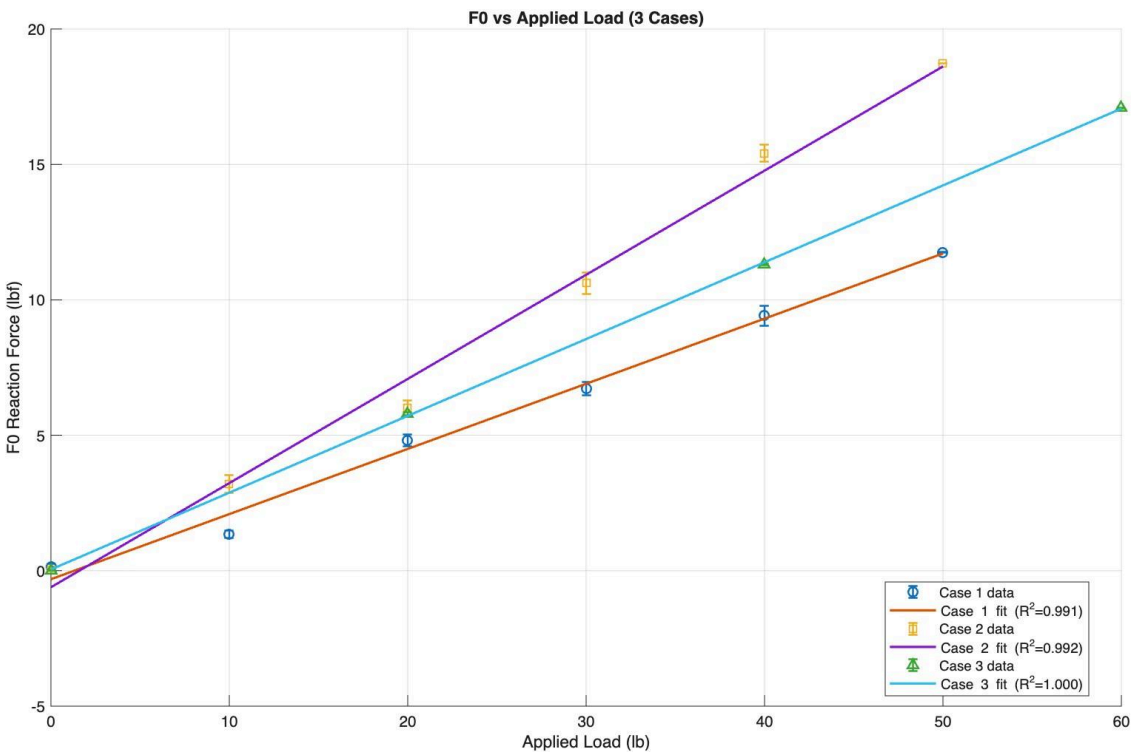
[1] Quinn Jochim, ASEN 3802, CU Boulder
[2] Anton Pylypchenko, ASEN 3802 CU Boulder
[3] Ritik Sarraf, ASEN 3802, CU Boulder
[4] Harroop Sooch, ASEN 3802, CU Boulder





[1] Quinn Jochim, ASEN 3802, CU Boulder
[2] Anton Pylypchenko, ASEN 3802 CU Boulder
[3] Ritik Sarraf, ASEN 3802, CU Boulder
[4] Harroop Sooch, ASEN 3802, CU Boulder





Part 2 task 1:

[1] Quinn Jochim, ASEN 3802, CU Boulder
[2] Anton Pylypchenko, ASEN 3802 CU Boulder
[3] Ritik Sarraf, ASEN 3802, CU Boulder
[4] Harroop Sooch, ASEN 3802, CU Boulder

	F0	F1	F2	F3D	LVDT
Case 1	0.99096	0.99577	0.99979	0.99823	0.99018
Case 2	0.99211	0.98468	0.99977	0.99809	0.87623
Case 3	0.99991	0.99986	1	0.99961	0.99981

As can be seen in the LVDT deflection versus applied load plot, a clear positive linear relationship between displacement and applied load is observed. The remaining plots for the reaction forces (F0, F1, F2) and the internal force at the center of the truss (F3D) exhibit the same highly linear behavior, indicating that both deflection and internal forces scale proportionally with load. This response is consistent with the truss operating within the linear-elastic regime. Minor deviations from perfect linearity are attributed to experimental scatter, including sensor noise, small variations in load placement, and compliance in the supports.

To quantify this behavior, a least-squares linear regression was applied to each measured quantity as a function of applied load for all three cases. Measurements taken at identical load levels were first averaged to reduce random error, and a first-order model was then fit to the averaged data. The quality of each fit was assessed using the coefficient of determination, R^2, with values close to unity indicating strong agreement between the experimental data and the linear model. The consistently high R^2 values across all sensors and loading cases confirm the validity of the linear assumption and demonstrate repeatable, elastic structural behavior throughout the experiment.

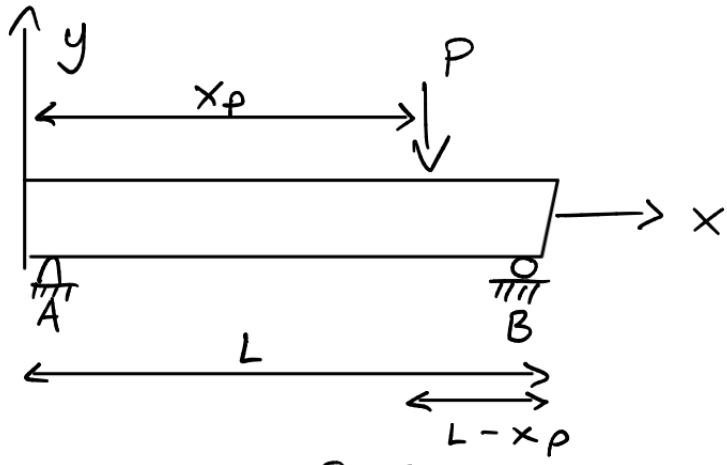
Part 2 task 2:

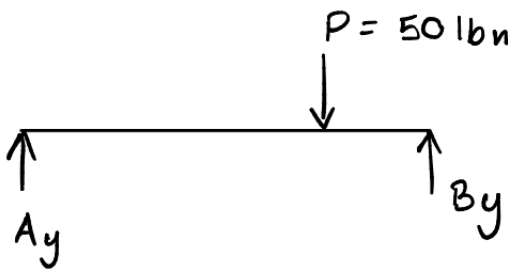
Can you identify the position of the loading, based on the reaction forces? Why?

Yes, when a single load P is applied at position xp (from left) The reaction forces (AY and BY) become functions of P, L and xp and since P and L are known we can solve for xp.

[1] Quinn Jochim, ASEN 3802, CU Boulder
[2] Anton Pylypchenko, ASEN 3802 CU Boulder
[3] Ritik Sarraf, ASEN 3802, CU Boulder
[4] Harroop Sooch, ASEN 3802, CU Boulder

Part 2 task 2





$$\Sigma M_A = 0 = -P(x_p) + B_y(L)$$

$$\rightarrow x_p = \frac{B_y L}{P}$$

$$\Sigma M_B = 0 = -A_y(L) + P(L - x_p)$$

$$\rightarrow x_p = L\left(1 - \frac{A_y}{P}\right)$$

[1] Quinn Jochim, ASEN 3802, CU Boulder
[2] Anton Pylypchenko, ASEN 3802 CU Boulder
[3] Ritik Sarraf, ASEN 3802, CU Boulder
[4] Harroop Sooch, ASEN 3802, CU Boulder

$$x_p = \frac{B_y L}{P} = L(1 - \frac{Ay}{P})$$

Where: $A_y = F0 + F1$ and $B_y = F2$

The reaction forces were first averaged then the two equations were solved yielding a value of xp = 5.0776 ft (from the By equation) and xp = 4.5145 ft (from the Ay equation) these values were then averaged to a value of xp = 4.7960 ft, this value was used in conjunction with the equivalent beam model equation for case 2 to calculate a predicted displacement.

using $v(L/2) = \frac{Pa}{24EI}(3L^2 - 4a^2)$

at $a = x_{p_{avg}} = 4.7960\,ft = 57.552\,in$

$I = 5.94\,in^4$

$L = 4m = 157.4\,in$

$P = 50\,lbm$

$E = 69\,GPa = 10 \cdot 10^6\,psi$

$$v\left(\frac{L}{2}\right) = \frac{50(57.552)}{24(10 \cdot 10^6)(5.94)}\left(3\left(\frac{157.4}{2}\right)^2 - 4(57.552)^2\right) = \boxed{.011\,in}$$

Equivalent beam model displacement

The actual displacement was notably larger with a magnitude of .04 in (this was done in matlab where only the points corresponding to 50 lbs were averaged)

Now comparing the internal force of the experimental vs predicted models. For the predicted value I used the equivalent beam model equation for stress with the equation for internal force.

[1] Quinn Jochim, ASEN 3802, CU Boulder
[2] Anton Pylypchenko, ASEN 3802 CU Boulder
[3] Ritik Sarraf, ASEN 3802, CU Boulder
[4] Harroop Sooch, ASEN 3802, CU Boulder

Using $\sigma = \frac{Pac}{I}$ and $F_i = \sigma A$

$c = 4.90\,in$

at $a = x_{p_{avg}} = 4.7960\,ft = 57.552\,in$

$I = 5.94\,in^4 \quad A = .06136\,in^2$

$$\sigma = \frac{50(57.552)(4.90)}{5.94} = 2373.78$$

$$F_i = 2373.78(.06136) = 145.66\,lbf$$

Then for the experimental data, the internal force near midspan was obtained by subtracting the average unloaded F3D reading from the average F3D reading at full load, where both values were computed by averaging measurements within the selected load ranges to remove sensor offset and measurement noise. This yielded F_i = 51.303 lbf again this marks a noticeable difference between the modeled data and experimental data.

[1] Quinn Jochim, ASEN 3802, CU Boulder
[2] Anton Pylypchenko, ASEN 3802 CU Boulder
[3] Ritik Sarraf, ASEN 3802, CU Boulder
[4] Harroop Sooch, ASEN 3802, CU Boulder

PART 2 TASK 3:

Part 2 Task 3

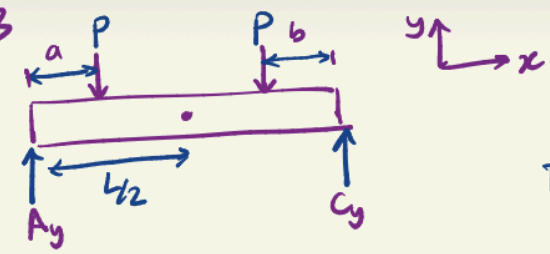
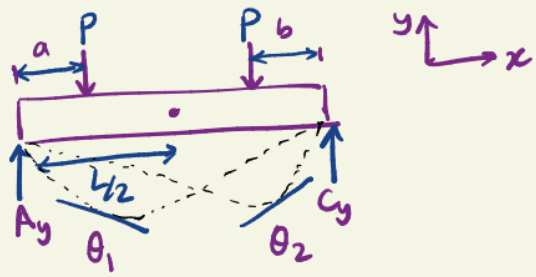


figure 2.3(i)



$$\boxed{\theta_1 = \theta_2}$$

figure 2.3(ii)

Consider a cantilever beam supported by two symmetric loads on either side at the same distance from its edge as shown on the figure

Givens:

$A_y = C_y$

$a = b$

$I = 5.94 \text{ in}^4$

$L = 4m = 157.4 \text{ in}$

$P = 50 \text{ lbm}$

$E = 69 \text{ GPa} \approx 10 \times 10^6 \text{ psi}$

[1] Quinn Jochim, ASEN 3802, CU Boulder
[2] Anton Pylypchenko, ASEN 3802 CU Boulder
[3] Ritik Sarraf, ASEN 3802, CU Boulder
[4] Harroop Sooch, ASEN 3802, CU Boulder

$$C_1 = -\frac{PL^2}{8}$$

$$= \frac{-(50\,lbm)(157.4)^2}{8}$$

$$= -1.548 \times 10^5 \,(lbm \cdot in^2)$$

$$\frac{Px^2}{2} + C_1 = 0$$

$$\frac{(50)\,x^2}{2} + (-1.548 \times 10^5) = 0$$

$$\boxed{x = 78.7 \text{ in}}$$

For left section (Symmetric load)

$$v(x) = \frac{P}{6EI}\left(x^3 - \frac{3L^2}{4}x\right)$$

[1] Quinn Jochim, ASEN 3802, CU Boulder
[2] Anton Pylypchenko, ASEN 3802 CU Boulder
[3] Ritik Sarraf, ASEN 3802, CU Boulder
[4] Harroop Sooch, ASEN 3802, CU Boulder

$$v(x) = \frac{(50\,lbm)}{6(10\times10^6\,psi)(5.94)}\left[(78.7)^3 - \frac{3(157.4)^2}{4}(78.7)\right]$$

$$= \frac{50\,lbm}{6(10\times10^6)(5.94)}\left[-9.7489\times10^5\right]$$

$$= -1.368\times10^{-1}$$

$$= -0.1368\ in$$

For Right section (Symmetric Load)

$$v(x) = \frac{P(L-x)}{6EI}\left(2aL - a^2 - (L-x)^2\right)$$

$$= \frac{50(157.4-78.7)}{6(10\times10^6)(5.94)}\left(2(57.552)(157.4) - (57.552)^2 - (157.4-57.552)^2\right)$$

[1] Quinn Jochim, ASEN 3802, CU Boulder
[2] Anton Pylypchenko, ASEN 3802 CU Boulder
[3] Ritik Sarraf, ASEN 3802, CU Boulder
[4] Harroop Sooch, ASEN 3802, CU Boulder

$$= \frac{3.935\times10^{3}}{3.564\times10^{8}}\left(4.836\times10^{3}\right)$$

$$= 5.339\times10^{-2}\ \text{in}$$

$$\boxed{= 0.05339\ \text{in}}$$

Discussion

The task 3 specifically asks for the loading at either side of the cantilever beam where the equal and symmetric loads are applied in order to approximate the deflection. When the loads are symmetric and placed equidistant from each end of the beam. The magnitude of the slope of the deflection at that point would be the same and will mirror itself as shown on the above figure.

Yes, when a single load P is applied at position to the left and right portion. The reaction forces (AY and BY) become functions of P, L and x_p and since P and L are known we can solve for x_p.

The deflection for the left portion with negative value but still here we are considering just the magnitude (0.1368 in) and shows downward deflection of the beam at that point using an equivalent beam model with symmetric loads. The right portion v(x) shows the deflection of 0.05339 in. As both (right and left) portions do not align to be the same value and this shows that the equivalent beam deflection model in this case is not valid.

[1] Quinn Jochim, ASEN 3802, CU Boulder
[2] Anton Pylypchenko, ASEN 3802 CU Boulder
[3] Ritik Sarraf, ASEN 3802, CU Boulder
[4] Harroop Sooch, ASEN 3802, CU Boulder

Contribution Report - Structural Mechanics

Ritik Sarraf: I did work on completing task 3**.**
Quinn Jochim: Completed all of tasks 1 and 2
Anton Pylypchenko: I did not contribute to part 2, have been working on part 3

Harroop Sooch:

Appendix

Code for Part 2 Task 1

```
clc; clear; close all;

% File names for each loading case
case1_file = 'case_1_data.txt';
case2_file = 'case_2_data.txt';
case3_file = 'case_3_data.txt';

% Load raw experimental data
C1 = readmatrix(case1_file);
C2 = readmatrix(case2_file);
C3 = readmatrix(case3_file);

% Map data columns to physical measurements
getCols = @(C) struct( ...
    'Load', C(:,1), ...
    'F0',   C(:,2), ...
    'F1',   C(:,3), ...
    'F2',   C(:,4), ...
    'F3D',  C(:,5), ...
    'LVDT', C(:,6));
```

[1] Quinn Jochim, ASEN 3802, CU Boulder
[2] Anton Pylypchenko, ASEN 3802 CU Boulder
[3] Ritik Sarraf, ASEN 3802, CU Boulder
[4] Harroop Sooch, ASEN 3802, CU Boulder

```
S1 = getCols(C1);
S2 = getCols(C2);
S3 = getCols(C3);

% Function handle to average repeated measurements at each load
avgByLoad = @(Load,y) localAvgByLoad(Load,y);

% Function handle for plotting all three cases with linear fits
plot3CasesWithFit = @(yname, ylab)
localPlot3CasesWithFit(S1,S2,S3,avgByLoad,yname,ylab);

% Generate required plots (measurement vs applied load)
plot3CasesWithFit('F0' , 'F0 Reaction Force (lbf)');
plot3CasesWithFit('F1' , 'F1 Reaction Force (lbf)');
plot3CasesWithFit('F2' , 'F2 Reaction Force (lbf)');
plot3CasesWithFit('F3D', 'Internal Force F3D (lbf)');
plot3CasesWithFit('LVDT','LVDT Displacement (in)');

% Print regression parameters and R^2 values
localPrintFitStats(S1,S2,S3,avgByLoad,'F0');
localPrintFitStats(S1,S2,S3,avgByLoad,'F1');
localPrintFitStats(S1,S2,S3,avgByLoad,'F2');
localPrintFitStats(S1,S2,S3,avgByLoad,'F3D');
localPrintFitStats(S1,S2,S3,avgByLoad,'LVDT');

% Compute R^2 summary table for report
fields    = {'F0','F1','F2','F3D','LVDT'};
caseNames = {'Case 1','Case 2','Case 3'};
cases     = {S1,S2,S3};

R2_mat = zeros(numel(cases), numel(fields));

for i = 1:numel(cases)
    for j = 1:numel(fields)
        [L, ymean] = avgByLoad(cases{i}.Load, cases{i}.(fields{j}));
        p = polyfit(L, ymean, 1);
        R2_mat(i,j) = localR2(ymean, polyval(p,L));
    end
end

R2_Table = array2table(R2_mat, ...
    'VariableNames', fields, ...
    'RowNames', caseNames);

disp('----- R^2 Summary Table -----');
disp(R2_Table);

% ================= LOCAL FUNCTIONS =================

function [L, y_mean, y_std] = localAvgByLoad(Load, y)
    L = unique(Load);
    y_mean = zeros(size(L));
```

[1] Quinn Jochim, ASEN 3802, CU Boulder
[2] Anton Pylypchenko, ASEN 3802 CU Boulder
[3] Ritik Sarraf, ASEN 3802, CU Boulder
[4] Harroop Sooch, ASEN 3802, CU Boulder

```
    y_std  = zeros(size(L));
    for i = 1:numel(L)
        idx = Load == L(i);
        y_mean(i) = mean(y(idx));
        y_std(i)  = std(y(idx));
    end
end

function localPlot3CasesWithFit(S1,S2,S3,avgByLoad,yfield,ylab)
    [L1, y1m, y1s] = avgByLoad(S1.Load, S1.(yfield));
    [L2, y2m, y2s] = avgByLoad(S2.Load, S2.(yfield));
    [L3, y3m, y3s] = avgByLoad(S3.Load, S3.(yfield));

    p1 = polyfit(L1, y1m, 1); y1fit = polyval(p1, L1); R21 = localR2(y1m,
y1fit);
    p2 = polyfit(L2, y2m, 1); y2fit = polyval(p2, L2); R22 = localR2(y2m,
y2fit);
    p3 = polyfit(L3, y3m, 1); y3fit = polyval(p3, L3); R23 = localR2(y3m,
y3fit);

    figure; hold on; grid on;

    errorbar(L1, y1m, y1s, 'o', 'LineWidth',1.2);
    plot(L1, y1fit, '-', 'LineWidth',1.5);

    errorbar(L2, y2m, y2s, 's', 'LineWidth',1.2);
    plot(L2, y2fit, '-', 'LineWidth',1.5);

    errorbar(L3, y3m, y3s, '^', 'LineWidth',1.2);
    plot(L3, y3fit, '-', 'LineWidth',1.5);

    xlabel('Applied Load (lb)');
    ylabel(ylab);
    title(sprintf('%s vs Applied Load (3 Cases)', yfield));

    legend( ...
        sprintf('Case 1 data'), sprintf('Case 1 fit (R^2=%.3f)',R21), ...
        sprintf('Case 2 data'), sprintf('Case 2 fit (R^2=%.3f)',R22), ...
        sprintf('Case 3 data'), sprintf('Case 3 fit (R^2=%.3f)',R23), ...
        'Location','best');
end

function R2 = localR2(y, yfit)
    SSres = sum((y - yfit).^2);
    SStot = sum((y - mean(y)).^2);
    R2 = 1 - SSres/SStot;
end

function localPrintFitStats(S1,S2,S3,avgByLoad,yfield)
    [L1, y1m] = avgByLoad(S1.Load, S1.(yfield));
    [L2, y2m] = avgByLoad(S2.Load, S2.(yfield));
    [L3, y3m] = avgByLoad(S3.Load, S3.(yfield));
```

```
    p1 = polyfit(L1,y1m,1); R21 = localR2(y1m,polyval(p1,L1));
    p2 = polyfit(L2,y2m,1); R22 = localR2(y2m,polyval(p2,L2));
    p3 = polyfit(L3,y3m,1); R23 = localR2(y3m,polyval(p3,L3));

    fprintf('%s: Case1 slope=%.4g, intercept=%.4g, R^2=%.4f | ', yfield,
p1(1), p1(2), R21);
    fprintf('Case2 slope=%.4g, intercept=%.4g, R^2=%.4f | ', yfield,
p2(1), p2(2), R22);
    fprintf('Case3 slope=%.4g, intercept=%.4g, R^2=%.4f\n', yfield, p3(1),
p3(2), R23);
End
```

Code for Part 2 task 2:

```
clc; clear; close all;
%% Inputs
case2_file = 'case_2_data.txt';
L_ft     = 13.12;   % span (ft)
P_target = 50;      % lbf
P_tol    = 2.0;     % lbf
%% Load data
C2 = readmatrix(case2_file);
Load = C2(:,1);
F0   = C2(:,2);
F1   = C2(:,3);
F2   = C2(:,4);
F3D  = C2(:,5);
LVDT = C2(:,6);
%% Average by load level
[L_unique, F0m]   = avgByLoad(Load, F0);
[~,        F1m]   = avgByLoad(Load, F1);
[~,        F2m]   = avgByLoad(Load, F2);
[~,        F3Dm]  = avgByLoad(Load, F3D);
[~,        LVDTm] = avgByLoad(Load, LVDT);
Ay_m = F0m + F1m;
By_m = F2m;
%% Select "full load" points
idx_full = abs(L_unique - P_target) <= P_tol;
if ~any(idx_full)
   error('No load levels within ±%.2f lbf of %.2f lbf.', P_tol, P_target);
end
P_full  = mean(L_unique(idx_full));
Ay_full = mean(Ay_m(idx_full));
By_full = mean(By_m(idx_full));
%% Load location estimates
xp1_ft = (By_full / P_full) * L_ft;
xp2_ft = L_ft * (1 - Ay_full / P_full);
xp_avg_ft = mean([xp1_ft, xp2_ft]);
%% Experimental midspan displacement at full load
v_mid_exp_avg = mean(LVDTm(idx_full));
v_mid_exp_std = std(LVDTm(idx_full));
%% Experimental internal force near midspan (ΔF3D)
[~, idx0] = min(abs(L_unique));
F3D_unloaded_avg = F3Dm(idx0);
```

[1] Quinn Jochim, ASEN 3802, CU Boulder
[2] Anton Pylypchenko, ASEN 3802 CU Boulder
[3] Ritik Sarraf, ASEN 3802, CU Boulder
[4] Harroop Sooch, ASEN 3802, CU Boulder

```
F3D_loaded_avg   = mean(F3Dm(idx_full));
F3D_loaded_std   = std(F3Dm(idx_full));
dF3D_avg = F3D_loaded_avg - F3D_unloaded_avg;
%% Print
fprintf('\n===== Case 2: Reaction-Based Load Location =====\n');
fprintf('Full-load points: %d (within ±%.2f lbf of %.2f lbf)\n', sum(idx_full),
P_tol, P_target);
fprintf('P_full  = %.4f lbf\nAy_full = %.4f lbf\nBy_full = %.4f lbf\n', P_full,
Ay_full, By_full);
fprintf('\nxp estimates:\n');
fprintf('xp1 = %.4f ft (from By)\n', xp1_ft);
fprintf('xp2 = %.4f ft (from Ay)\n', xp2_ft);
fprintf('xp  = %.4f ft = %.2f in = %.3f m\n', xp_avg_ft, xp_avg_ft*12,
xp_avg_ft/3.28084);
fprintf('\n===== Experimental Midspan Displacement (Full Load) =====\n');
fprintf('v_mid_exp_avg = %.5f in\nv_mid_exp_std = %.5f in\n', v_mid_exp_avg,
v_mid_exp_std);
fprintf('\n===== Experimental Internal Force Near Midspan (F3D) =====\n');
fprintf('F3D_unloaded_avg = %.5f lbf\n', F3D_unloaded_avg);
fprintf('F3D_loaded_avg   = %.5f lbf\nF3D_loaded_std   = %.5f lbf\n',
F3D_loaded_avg, F3D_loaded_std);
fprintf('ΔF3D_avg         = %.5f lbf\n', dF3D_avg);
%% Local function
function [L, y_mean] = avgByLoad(Load, y)
L = unique(Load);
y_mean = zeros(size(L));
for i = 1:numel(L)
   idx = (Load == L(i));
   y_mean(i) = mean(y(idx));
end
end
```

[1] Quinn Jochim, ASEN 3802, CU Boulder
[2] Anton Pylypchenko, ASEN 3802 CU Boulder
[3] Ritik Sarraf, ASEN 3802, CU Boulder
[4] Harroop Sooch, ASEN 3802, CU Boulder